

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jan 2022 P2HR Worked Solutions

Jan 2022 | P2HR

27 questions ■ question image followed by worked solution

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STUDENT EDITION

Worked answers included

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PAST PAPER QUESTION**Q#: 1** **Marks: 4****TOPIC: Probability Toolkit**

Jan 2022 P2HR - Jan 2022 | P2HR

- 1** A tin contains tea bags with a choice of four different flavours of tea. The four flavours of tea are Assam or Darjeeling or Nilgiri or Rize.

Sara takes at random a tea bag from the tin.

The table shows each of the probabilities that the flavour of the tea Sara takes is Assam or Darjeeling or Rize.

Flavour of tea	Assam	Darjeeling	Nilgiri	Rize
Probability	0.38	0.24		0.16

- (a) Work out the probability that the flavour of the tea Sara takes is Nilgiri.

.....
(2)

- (b) Work out the probability that the flavour of the tea Sara takes is either Darjeeling or Rize.

.....
(2)

(Total for Question 1 is 4 marks)

PAST PAPER SOLUTION

Q#: 1 Marks: 4

TOPIC: **Probability Toolkit**

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WORKED SOLUTION

Dr. Eslam Ahmed

1. The missing probability is

The missing probability is

$$1 - 0.38 - 0.24 - 0.16 = 0.22$$

2. Calculate probability

So

$$P(\text{Darjeeling or Rize}) = 0.24 + 0.16 = 0.40$$

FINAL ANSWER

Nilgiri 0.22, Darjeeling or Rize 0.40.

PAST PAPER QUESTION**Q#: 2** **Marks: 7**TOPIC: **Percentages**

Jan 2022 P2HR - Jan 2022 | P2HR

2 Mary saves for a holiday each year.

In 2020 she saved a total of \$720

In 2021, each month she saved \$78

The total amount Mary saved in 2021 was $P\%$ more than the total she saved in 2020(a) Work out the value of P

(4)

Roberto is going to go on holiday.

He has two coupons that will save him money on his holiday.

Coupon A18% off the cost of the
accommodation**Coupon B**12.5% off the total cost of the
accommodation **and** the flights

For Roberto's holiday

the cost of the accommodation is \$1600

the cost of the flights is \$800

Roberto can only use one of the coupons.

He wants to save as much money as he can.

(b) Which of the two coupons, **A** or **B**, should he use?

Show your working clearly.

(3)

(Total for Question 2 is 7 marks)

PAST PAPER SOLUTION**Q#: 2****Marks: 7****TOPIC: Percentages**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Mary saved in 2021**

Mary saved in 2021:

$$78 \times 12 = 936$$

2. Increase from 2020

Increase from 2020:

$$936 - 720 = 216$$

3. Percentage increase

Percentage increase:

$$\frac{216}{720} \times 100 = 30$$

4. Calculate percentageSo $P = 30$.**5. Coupon A saves**

Coupon A saves

$$18\% \times 1600 = 0.18 \times 1600 = 288$$

6. Coupon B saves of the

Coupon B saves 12.5% of the accommodation and flights together:

$$1600 + 800 = 2400$$

$$12.5\% \times 2400 = 0.125 \times 2400 = 300$$

7. Coupon B saves more money

Coupon B saves more money.

FINAL ANSWER(a) $P = 30$, (b) Coupon B

PAST PAPER QUESTION**Q#: 3****Marks: 5****TOPIC: Solving Linear Equations**

Jan 2022 P2HR - Jan 2022 | P2HR

3 (a) Solve $4y + 5 > 12$
(2)(b) Solve $6x - 5 = \frac{4x - 7}{2}$

Show clear algebraic working.

 $x =$
(3)

(Total for Question 3 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 3****Marks: 5****TOPIC: Solving Linear Equations**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. (a)**

(a)

$$4y + 5 > 12$$

$$4y > 7$$

$$y > \frac{7}{4}$$

2. (b)

(b)

$$6x - 5 = \frac{4x - 7}{2}$$

3. Multiply by 2

Multiply by 2:

$$12x - 10 = 4x - 7$$

$$8x = 3$$

$$x = \frac{3}{8}$$

FINAL ANSWER

$$y > \frac{7}{4}, \text{ and } x = \frac{3}{8}.$$

PAST PAPER QUESTION**Q#: 4****Marks: 4****TOPIC: Angles in Polygons & Parallel Lines**

Jan 2022 P2HR - Jan 2022 | P2HR

- 4 The diagram shows a regular octagon $ABCDEFGH$ and a regular pentagon $ABIJK$

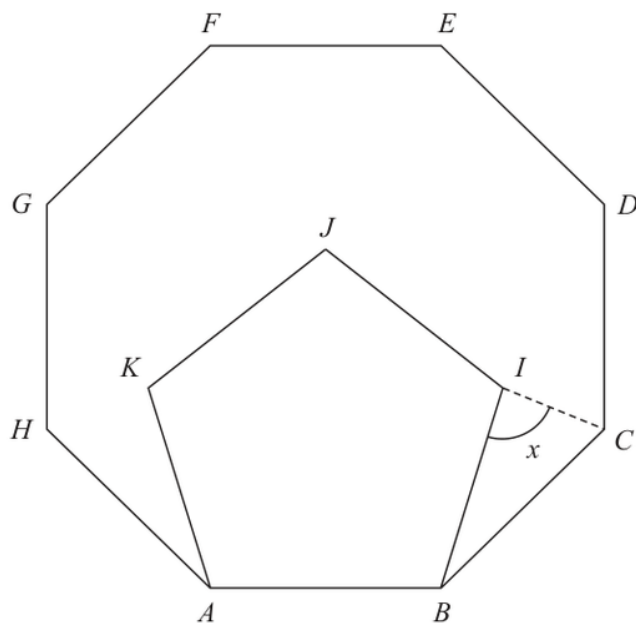


Diagram **NOT**
accurately drawn

Work out the size of the angle x

o

(Total for Question 4 is 4 marks)

PAST PAPER SOLUTION**Q#: 4****Marks: 4****TOPIC: Angles in Polygons & Parallel Lines**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The interior angle of a**

The interior angle of a regular octagon is

$$\frac{(8 - 2) \times 180}{8} = 135^\circ$$

2. The interior angle of a

The interior angle of a regular pentagon is

$$\frac{(5 - 2) \times 180}{5} = 108^\circ$$

3. Find the gradientAt B ,

$$\angle IBC = 135^\circ - 108^\circ = 27^\circ$$

4. Find the gradientSince both polygons share side AB ,

$$BI = BC$$

5. Find the gradientSo triangle BIC is isosceles.

$$x = \frac{180^\circ - 27^\circ}{2} = 76.5^\circ$$

FINAL ANSWER

76.5°.

PAST PAPER QUESTION**Q#: 5** **Marks: 3****TOPIC: Compound Interest & Depreciation**

Jan 2022 P2HR - Jan 2022 | P2HR

- 5** Shane invests 7200 dollars for 3 years in a savings account.
He gets 2.5% per year compound interest.

How much money will Shane have in his savings account at the end of 3 years?
Give your answer to the nearest dollar.

..... dollars

(Total for Question 5 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 5** **Marks: 3****TOPIC: Compound Interest & Depreciation**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The multiplier for compound interest**

The multiplier for 2.5% compound interest is

$$1.025$$

2. Use compound interest

After 3 years,

$$7200(1.025)^3 = 7753.6125$$

3. Use compound interest

Correct to the nearest dollar,

$$7753.6125 \approx 7754$$

FINAL ANSWER

7754 dollars

PAST PAPER QUESTION**Q#: 6** **Marks: 4**TOPIC: **Algebraic Roots & Indices**

Jan 2022 P2HR - Jan 2022 | P2HR

- 6 (a) Write down the value of x^0

.....
(1)

Given that $2^{-3} \times 2^9 = 2^n$

- (b) find the value of n

$n =$
(1)

Given that $\frac{7^{206} \times 7^m}{7^{214}} = 7^{-3}$

- (c) find the value of m

$m =$
(2)

(Total for Question 6 is 4 marks)

PAST PAPER SOLUTION**Q#: 6****Marks: 4****TOPIC: Algebraic Roots & Indices**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use index laws**

$$x^0 = 1$$

2. Use index laws

For part (b),

$$2^{-3} \times 2^9 = 2^{-3+9} = 2^6$$

3. Use index laws

So

$$n = 6$$

4. Use index laws

For part (c),

$$\frac{7^{206} \times 7^m}{7^{214}} = 7^{206+m-214}$$

5. Use index lawsThis equals 7^{-3} , so

$$206 + m - 214 = -3$$

$$m = 5$$

FINAL ANSWER(a) 1, (b) $n = 6$, (c) $m = 5$.

PAST PAPER QUESTION**Q#: 7** **Marks: 5****TOPIC: Graphing Inequalities**

Jan 2022 P2HR - Jan 2022 | P2HR

- 7 (a) Write down an equation of the straight line with gradient -3 and which passes through the point with coordinates $(0, 5)$

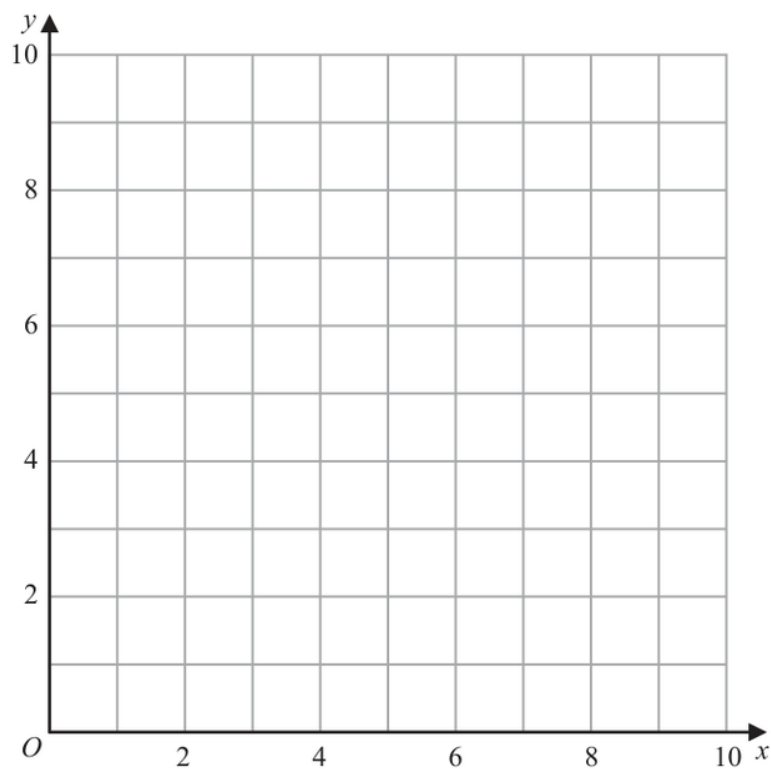
(2)

- (b) Show, by shading on the grid, the region defined by **all three** of the inequalities

$x \leq 6$

$y \geq 2$

$y \leq x + 1$

Label the region **R**

(3)

(Total for Question 7 is 5 marks)

PAST PAPER SOLUTION**Q#: 7****Marks: 5****TOPIC: Graphing Inequalities**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. A straight line with gradient**

A straight line with gradient -3 passing through $(0, 5)$ has equation

$$y = -3x + 5$$

2. Use the inequalities

For the region, use the inequalities:

$$x < 6$$

$$y > 2$$

$$y \leq x + 1$$

FINAL ANSWER

(a) $y = -3x + 5$. (b) Shade $x < 6$, $y > 2$, $y \leq x + 1$.

PAST PAPER QUESTION**Q#: 8****Marks: 3****TOPIC: Statistics Toolkit**

Jan 2022 P2HR - Jan 2022 | P2HR

- 8** A scientist is investigating the weight of 50 tigers.

Here is some information about these tigers.

	Type of tiger	
	Siberian	Bengal
Number of tigers	22	28
Mean weight of tigers (kg)	260	

The mean weight of all 50 tigers is 218 kg

Work out the mean weight of the Bengal tigers.

..... kg

(Total for Question 8 is 3 marks)

PAST PAPER SOLUTION**Q#: 8****Marks: 3****TOPIC: Statistics Toolkit**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Total mass of all 50**

Total mass of all 50 tigers:

$$50 \times 218 = 10900$$

2. Total mass of the 22

Total mass of the 22 Siberian tigers:

$$22 \times 260 = 5720$$

3. Total mass of the 28

Total mass of the 28 Bengal tigers:

$$10900 - 5720 = 5180$$

4. Mean mass of the Bengal

Mean mass of the Bengal tigers:

$$\frac{5180}{28} = 185$$

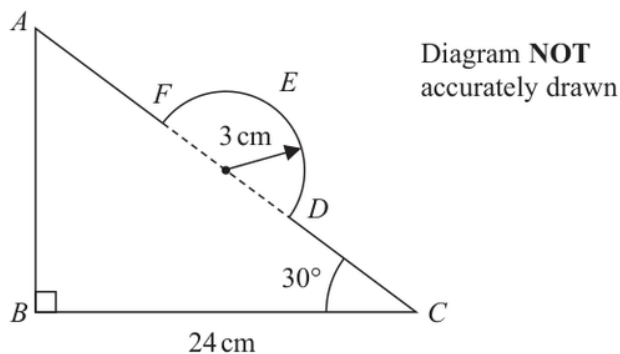
FINAL ANSWER

185 kg.

PAST PAPER QUESTION**Q#: 9** **Marks: 5****TOPIC: Circles, Arcs & Sectors**

Jan 2022 P2HR - Jan 2022 | P2HR

- 9 In the diagram, ABC is a right-angled triangle and DEF is a semicircular arc.



In triangle ABC

$$BC = 24 \text{ cm}$$

$$\text{angle } ABC = 90^\circ$$

$$\text{angle } BCA = 30^\circ$$

The points D and F lie on AC so that DF is the diameter of the semicircular arc DEF .
The radius of the semicircular arc is 3 cm.

Work out the length of $AFEDC$

Give your answer correct to 2 significant figures.

..... cm

(Total for Question 9 is 5 marks)

PAST PAPER SOLUTION

Q#: 9

Marks: 5

TOPIC: **Circles, Arcs & Sectors**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. Use trigonometryIn triangle ABC ,

$$\cos 30^\circ = \frac{24}{AC}$$

$$AC = \frac{24}{\cos 30^\circ} = 27.712\dots$$

2. Calculate value

The semicircle has radius 3 cm, so its diameter is 6 cm.

$$AF + DC = AC - DF = 27.712\dots - 6$$

3. Arc is a semicircleArc FED is a semicircle:

$$\text{arc length} = 3\pi$$

4. Calculate value

Therefore

$$AFEDC = 27.712\dots - 6 + 3\pi = 31.137\dots$$

FINAL ANSWER

31 cm.

PAST PAPER QUESTION**Q#: 10****Marks: 3****TOPIC: Powers, Roots & Standard Form**

Jan 2022 P2HR - Jan 2022 | P2HR

- 10** The table gives information about the population and the total amount of money, in dollars, spent on healthcare for two countries in 2016

Country	Total population	Total spent on healthcare (\$)
Austria	8.7×10^6	4.2×10^{10}
Luxembourg	6.3×10^5	3.7×10^9

Work out how much more was spent **per person** on healthcare in Luxembourg than in Austria.

Give your answer correct to the nearest whole number.

..... dollars

(Total for Question 10 is 3 marks)

PAST PAPER SOLUTION**Q#: 10** **Marks: 3****TOPIC: Powers, Roots & Standard Form**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Austria spending per person**

Austria spending per person:

$$\frac{4.2 \times 10^{10}}{8.7 \times 10^6} = 4827.586 \dots$$

2. Luxembourg spending per person

Luxembourg spending per person:

$$\frac{3.7 \times 10^9}{6.3 \times 10^5} = 5873.015 \dots$$

3. Difference

Difference:

$$5873.015 \dots - 4827.586 \dots = 1045.429 \dots$$

4. Convert standard form

To the nearest whole number, this is

1045

FINAL ANSWER

1045 dollars

PAST PAPER QUESTION**Q#: 11** **Marks: 4****TOPIC: Graphs of Functions**

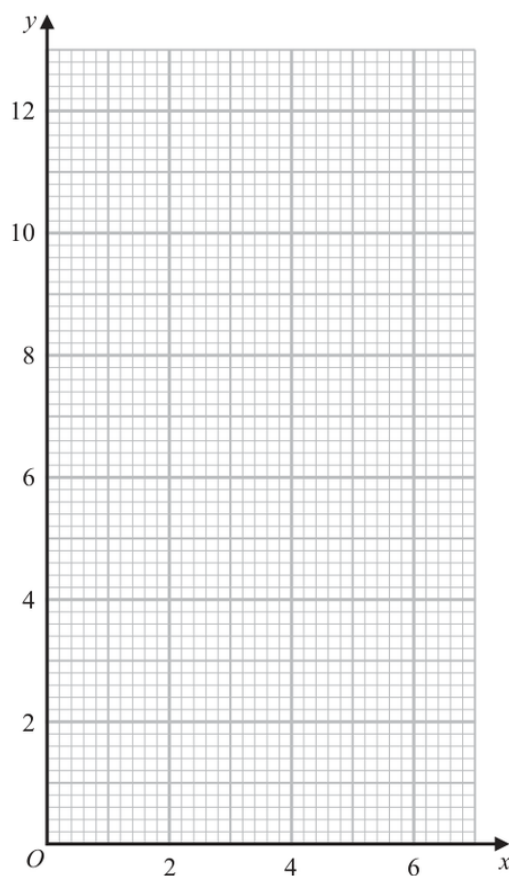
Jan 2022 P2HR - Jan 2022 | P2HR

- 11 (a) Complete the table of values for $y = \frac{6}{x}$

x	0.5	1	2	3	4	5	6
y		6		2			1

(2)

- (b) On the grid, draw the graph of $y = \frac{6}{x}$ for $0.5 \leq x \leq 6$



(2)

(Total for Question 11 is 4 marks)

PAST PAPER SOLUTION**Q#: 11** **Marks: 4****TOPIC: Graphs of Functions**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Read the graph**

$$y = \frac{6}{x}$$

2. The completed table is

The completed table is:

x	0.5	1	2	3	4	5	6
y	12	6	3	2	1.5	1.2	1

3. Plot the points and join

Plot the points and join them with a smooth decreasing reciprocal curve.

FINAL ANSWER

Missing values are 12, 3, 1.5, 1.2.

PAST PAPER QUESTION**Q#: 12****Marks: 3****TOPIC: Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2022 P2HR - Jan 2022 | P2HR

12 The diagram shows two vertical phone masts, AB and CD , on horizontal ground.

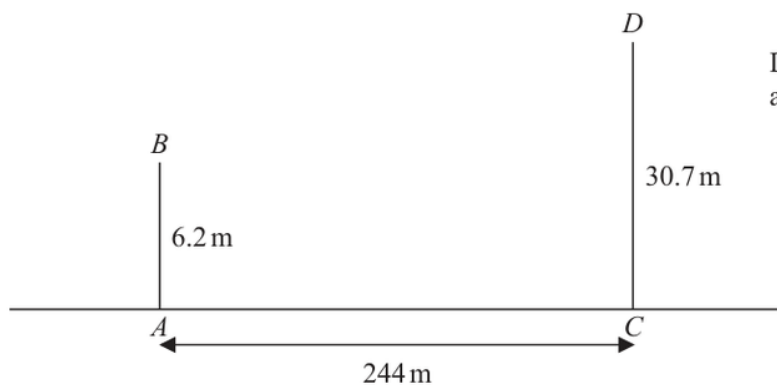


Diagram **NOT**
accurately drawn

$$AB = 6.2\text{ m} \quad AC = 244\text{ m} \quad CD = 30.7\text{ m}$$

Work out the size of the angle of depression of B from D
Give your answer correct to one decimal place.

o

(Total for Question 12 is 3 marks)

PAST PAPER SOLUTION

Q#: 12

Marks: 3

TOPIC: **Right-Angled Triangles - Pythagoras & Trigonometry**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. The vertical difference between the

The vertical difference between the tops of the masts is

$$30.7 - 6.2 = 24.5 \text{ m}$$

2. The horizontal distance is

The horizontal distance is 244 m.

$$\tan \theta = \frac{24.5}{244}$$

$$\theta = 5.733 \dots^\circ$$

FINAL ANSWER

5.7°.

PAST PAPER QUESTION**Q#: 13****Marks: 3****TOPIC: Algebraic Proof**

Jan 2022 P2HR - Jan 2022 | P2HR

13 $a = \sqrt{8} + 4$

$b = \sqrt{8} - 4$

 $(a - b)(a + b)$ can be written in the form $y\sqrt{4y}$ Find the value of y

Show your working clearly.

$y = \dots\dots\dots$

(Total for Question 13 is 3 marks)

EA

PAST PAPER SOLUTION**Q#: 13****Marks: 3****TOPIC: Algebraic Proof**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Simplify surd**

$$a = \sqrt{8} + 4, \quad b = \sqrt{8} - 4$$

$$a - b = (\sqrt{8} + 4) - (\sqrt{8} - 4) = 8$$

$$a + b = (\sqrt{8} + 4) + (\sqrt{8} - 4) = 2\sqrt{8} = 4\sqrt{2}$$

2. Simplify surd

So

$$(a - b)(a + b) = 8 \times 4\sqrt{2} = 32\sqrt{2}$$

3. The question says this equalsThe question says this equals $y\sqrt{4y}$. Try $y = 8$:

$$8\sqrt{32} = 8 \times 4\sqrt{2} = 32\sqrt{2}$$

FINAL ANSWER

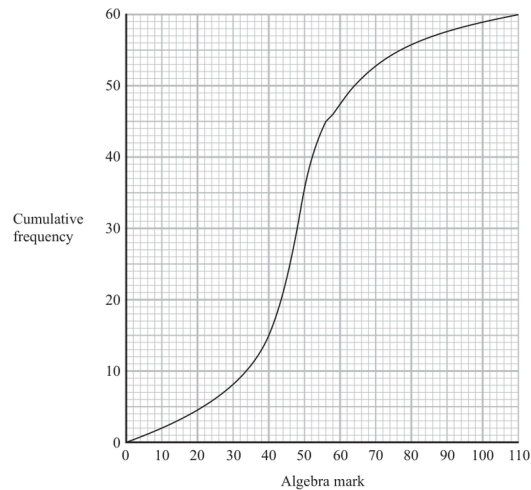
$$y = 8.$$

PAST PAPER QUESTION**Q#: 14****Marks: 8****TOPIC: Cumulative Frequency Diagrams**

Jan 2022 P2HR - Jan 2022 | P2HR

- 14** A group of 60 students each sat an algebra test and a geometry test. Each test was marked out of 110

The cumulative frequency graph gives information about the marks gained by the 60 students in the algebra test.



- (a) Use the graph to find an estimate for the median mark in the algebra test.

(1)

- (b) Use the graph to find an estimate for the number of students who gained 58 marks or less in the algebra test.

(1)

- (c) Use the graph to find an estimate for the interquartile range of the marks gained in the algebra test.

(2)

The interquartile range of the marks gained in the geometry test is 9

Luis says

“The students’ marks are more spread out in the algebra test than in the geometry test.”

- (d) Is Luis correct?
Give a reason for your answer.

(1)

To be awarded a grade A in the algebra test, a student had to gain a mark greater than 64

Two students are to be selected at random from the 60 students in the group.

- (e) Use the graph to find an estimate for the probability that both of these students were awarded a grade A in the algebra test.

(3)

(Total for Question 14 is 8 marks)

PAST PAPER SOLUTION**Q#: 14****Marks: 8****TOPIC: Cumulative Frequency Diagrams**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. There are students**

There are 60 students.

2. The median is at cumulative

The median is at cumulative frequency 30. From the graph,

$$\text{median} \approx 49$$

3. Use cumulative frequency

At 58 marks, the cumulative frequency is about 45, so about 45 students gained 58 marks or less.

4. Use cumulative frequency

For the interquartile range:

$$Q_1 \text{ is at } 15, \quad Q_3 \text{ is at } 45$$

5. Use cumulative frequency

From the graph,

$$Q_1 \approx 40, \quad Q_3 \approx 57$$

$$\text{IQR} \approx 57 - 40 = 17$$

6. Use cumulative frequency

The geometry IQR is 9, so Luis is correct because the algebra IQR is larger.

7. Use cumulative frequency

For grade A, use mark 64. From the graph, about 50 students gained 64 or less, so about

$$60 - 50 = 10$$

8. students gained more than

students gained more than 64.

9. Probability both selected students were

Probability both selected students were grade A:

$$\frac{10}{60} \times \frac{9}{59} = 0.0254 \dots$$

FINAL ANSWER

median about 49, about 45 students, IQR about 17, Luis is correct, probability about 0.025.

PAST PAPER QUESTION**Q#: 15****Marks: 4****TOPIC: Rearranging Formulas**

Jan 2022 P2HR - Jan 2022 | P2HR

15 Make t the subject of $n^2 = \frac{4d+t^3}{t^3}$

(Total for Question 15 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 15****Marks: 4****TOPIC: Rearranging Formulas**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Rearrange formula**

$$n^2 = \frac{4d + t^3}{t^3}$$

$$n^2 t^3 = 4d + t^3$$

$$t^3(n^2 - 1) = 4d$$

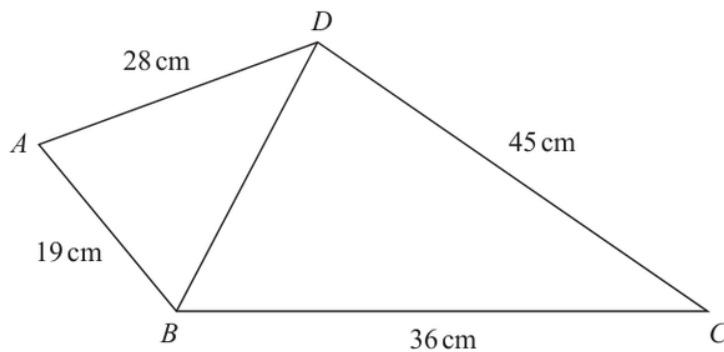
$$t^3 = \frac{4d}{n^2 - 1}$$

FINAL ANSWER

$$t = \sqrt[3]{\frac{4d}{n^2 - 1}}$$

PAST PAPER QUESTION**Q#: 16****Marks: 5****TOPIC: Sine, Cosine Rule & Area of Triangles**

Jan 2022 P2HR - Jan 2022 | P2HR

16 The diagram shows quadrilateral $ABCD$ Diagram **NOT**
accurately drawnThe angle BCD is acute.Given that the area of triangle $BCD = 405 \text{ cm}^2$ work out the size of angle ABD

Give your answer correct to one decimal place.

o

(Total for Question 16 is 5 marks)

PAST PAPER SOLUTION**Q#: 16****Marks: 5****TOPIC: Sine, Cosine Rule & Area of Triangles**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Use trigonometry**For triangle BCD ,

$$405 = \frac{1}{2}(36)(45) \sin BCD$$

$$\sin BCD = 0.5$$

2. Use trigonometryThe angle BCD is acute, so

$$BCD = 30^\circ$$

3. FindFind BD :

$$BD^2 = 36^2 + 45^2 - 2(36)(45) \cos 30^\circ$$

$$BD = 22.695 \dots$$

4. Use trigonometryNow use cosine rule in triangle ABD :

$$\cos ABD = \frac{19^2 + BD^2 - 28^2}{2(19)(BD)}$$

$$ABD = 83.871 \dots^\circ$$

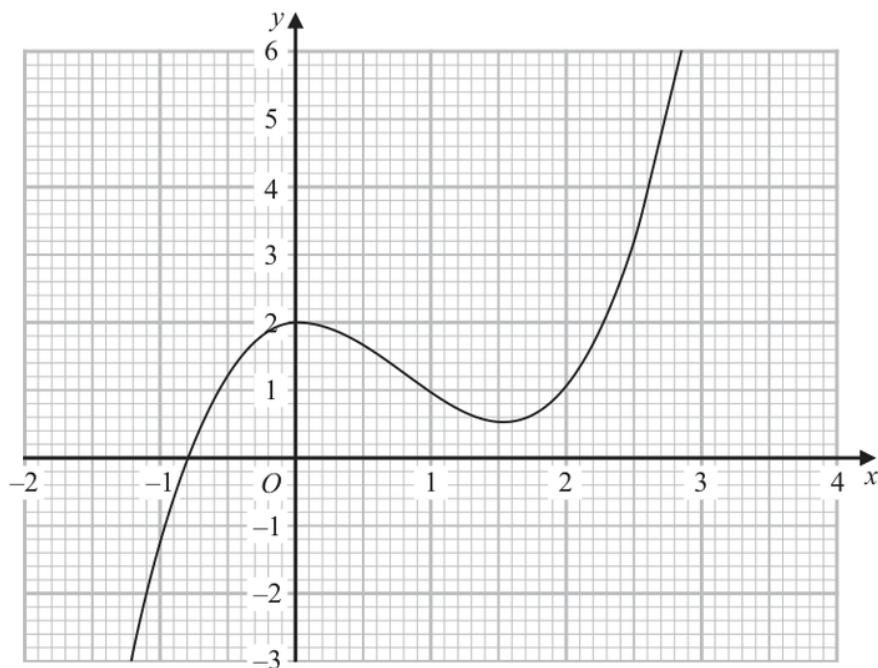
FINAL ANSWER

83.9°.

PAST PAPER QUESTION**Q#: 17****Marks: 3****TOPIC: Estimating Gradients**

Jan 2022 P2HR - Jan 2022 | P2HR

17 Part of the curve with equation $y = f(x)$ is shown on the grid.



Find an estimate for the gradient of the curve at the point where $x = 2$
Show your working clearly.

(Total for Question 17 is 3 marks)

PAST PAPER SOLUTION**Q#: 17** **Marks: 3****TOPIC: Estimating Gradients**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Draw a tangent to the**

Draw a tangent to the curve at the point where

$$x = 2$$

2. Find the gradient

Using two clear points on the tangent, for example approximately (1.5, 0.6) and (2.3, 2.2),

$$\text{gradient} \approx \frac{2.2 - 0.6}{2.3 - 1.5}$$

$$\text{gradient} \approx 2.0$$

FINAL ANSWER

The gradient is approximately 2.

PAST PAPER QUESTION**Q#: 18****Marks: 5****TOPIC: Coordinate Geometry**

Jan 2022 P2HR - Jan 2022 | P2HR

- 18** The line with equation $2y = x + 1$ intersects the curve with equation $3y^2 + 7y + 16 = x^2 - x$ at the points A and B

Find the coordinates of A and the coordinates of B
Show clear algebraic working.

(..... ,) and (..... ,)

(Total for Question 18 is 5 marks)

EA

PAST PAPER SOLUTION**Q#: 18****Marks: 5****TOPIC: Coordinate Geometry**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The line is**

The line is

$$2y = x + 1$$

2. Calculate value

So

$$x = 2y - 1$$

3. Substitute into the curve

Substitute into the curve

$$3y^2 + 7y + 16 = x^2 - x$$

$$3y^2 + 7y + 16 = (2y - 1)^2 - (2y - 1)$$

$$3y^2 + 7y + 16 = 4y^2 - 6y + 2$$

$$y^2 - 13y - 14 = 0$$

$$(y - 14)(y + 1) = 0$$

4. Calculate value

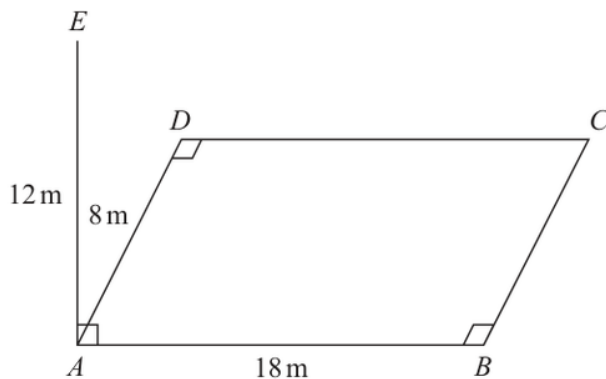
So

$$y = 14 \quad \text{or} \quad y = -1$$

5. Calculate valueIf $y = 14$, then $x = 27$.**6. Calculate value**If $y = -1$, then $x = -3$.**FINAL ANSWER** $(27, 14)$ and $(-3, -1)$.

PAST PAPER QUESTION**Q#: 19** **Marks: 3****TOPIC: 3D Pythagoras & Trigonometry**

Jan 2022 P2HR - Jan 2022 | P2HR

19 $ABCD$ is a horizontal rectangular field.Diagram **NOT**
accurately drawnA vertical pole, AE , is placed at the corner A of the field.

$$AE = 12\text{ m} \quad AB = 18\text{ m} \quad AD = 8\text{ m}$$

Calculate the size of the angle between EC and the plane $ABCD$

Give your answer correct to one decimal place.

o

(Total for Question 19 is 3 marks)

PAST PAPER SOLUTION

Q#: 19 Marks: 3

TOPIC: **3D Pythagoras & Trigonometry**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION

Dr. Eslam Ahmed

1. The projection of onto theThe projection of EC onto the field is AC .

$$AC = \sqrt{18^2 + 8^2} = \sqrt{388}$$

2. Use trigonometryThe vertical height is $AE = 12$ m, so

$$\tan \theta = \frac{12}{\sqrt{388}}$$

$$\theta = 31.350 \dots^\circ$$

FINAL ANSWER

31.4°.

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Direct & Inverse Proportion**

Jan 2022 P2HR - Jan 2022 | P2HR

20 y is inversely proportional to $\sqrt{x}$
 x is directly proportional to T^3

Given that $y = 8$ when $T = 25$

find the exact value of T when $y = 27$

$T = \dots\dots\dots$

(Total for Question 20 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 20****Marks: 4****TOPIC: Direct & Inverse Proportion**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find inverse function** y is inversely proportional to $\sqrt{x}$, so

$$y \propto \frac{1}{\sqrt{x}}$$

2. Find inverse function x is directly proportional to T^3 , so $\sqrt{x} \propto T^{3/2}$.**3. Find inverse function**

Therefore

$$y \propto \frac{1}{T^{3/2}}$$

4. Find inverse function

So

$$yT^{3/2} = k$$

5. Use whenUse $y = 8$ when $T = 25$:

$$k = 8 \times 25^{3/2} = 8 \times 125 = 1000$$

6. Find inverse functionWhen $y = 27$,

$$27T^{3/2} = 1000$$

$$T^{3/2} = \frac{1000}{27} = \left(\frac{10}{3}\right)^3$$

7. Raise both sides to theRaise both sides to the power $\frac{2}{3}$:

$$T = \left(\frac{10}{3}\right)^2 = \frac{100}{9}$$

FINAL ANSWER

$$T = \frac{100}{9}.$$

PAST PAPER QUESTION**Q#: 20****Marks: 4****TOPIC: Direct & Inverse Proportion**

Jan 2022 P2HR - Jan 2022 | P2HR

20 y is inversely proportional to $\sqrt{x}$
 x is directly proportional to T^3

Given that $y = 8$ when $T = 25$

find the exact value of T when $y = 27$

$T = \dots\dots\dots$

(Total for Question 20 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 20****Marks: 4****TOPIC: Direct & Inverse Proportion**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Find inverse function**

$$y \propto \frac{1}{\sqrt{x}}$$

2. Use Direct Inverse Proportion correct

and

$$x \propto T^3$$

3. Find inverse function

So

$$\sqrt{x} \propto T^{3/2}$$

4. Find inverse function

Therefore

$$y \propto \frac{1}{T^{3/2}}$$

5. Use Direct Inverse Proportion correctSo $yT^{3/2}$ is constant.**6. Find inverse function**When $y = 8$ and $T = 25$,

$$8(25)^{3/2} = 8(125) = 1000$$

7. Find inverse functionWhen $y = 27$,

$$27T^{3/2} = 1000$$

$$T^{3/2} = \frac{1000}{27} = \left(\frac{10}{3}\right)^3$$

$$T = \left(\frac{10}{3}\right)^2 = \frac{100}{9}$$

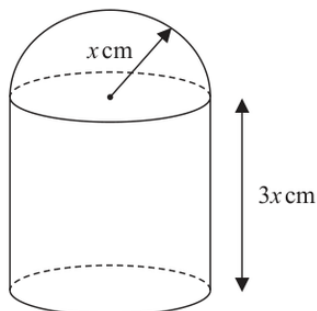
FINAL ANSWER

$$\frac{100}{9}$$

PAST PAPER QUESTION**Q#: 21****Marks: 6****TOPIC: Standard & Compound Units**

Jan 2022 P2HR - Jan 2022 | P2HR

- 21 The diagram shows a solid made from a cylinder and a hemisphere.
The cylinder and the hemisphere are both made from the same metal.

Diagram **NOT**
accurately drawn

The plane face of the hemisphere coincides with the upper plane face of the cylinder.

The radius of the cylinder and the radius of the hemisphere are both x cm.

The height of the cylinder is $3x$ cm.

The total surface area of the solid is $81\pi\text{cm}^2$

The mass of the solid is 840 grams.

The following table gives the density of each of four metals.

Metal	Density (g/cm^3)
Aluminium	2.7
Nickel	8.9
Gold	19.3
Silver	10.5

The metal used to make the solid is one of the metals in the table.

Determine the metal used to make the solid.

Show your working clearly.

(Total for Question 21 is 6 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 6****TOPIC: Standard & Compound Units**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The outside surface area is**

The outside surface area is:

2. curved surface area of cylinder

curved surface area of cylinder:

$$2\pi x(3x) = 6\pi x^2$$

3. bottom of cylinder

bottom of cylinder:

$$\pi x^2$$

4. curved surface area of hemisphere

curved surface area of hemisphere:

$$2\pi x^2$$

5. Calculate value

So

$$6\pi x^2 + \pi x^2 + 2\pi x^2 = 81\pi$$

$$9\pi x^2 = 81\pi$$

$$x^2 = 9$$

$$x = 3$$

6. Volume of the cylinder

Volume of the cylinder:

$$\pi x^2(3x) = 3\pi x^3$$

7. Volume of the hemisphere

Volume of the hemisphere:

$$\frac{2}{3}\pi x^3$$

8. Total volume

Total volume:

$$3\pi(3)^3 + \frac{2}{3}\pi(3)^3$$
$$= 81\pi + 18\pi = 99\pi$$

9. Density

Density:

$$\frac{840}{99\pi} = 2.70\dots$$

10. Compare density

This matches aluminium, whose density is 2.7 g/cm^3 .

FINAL ANSWER

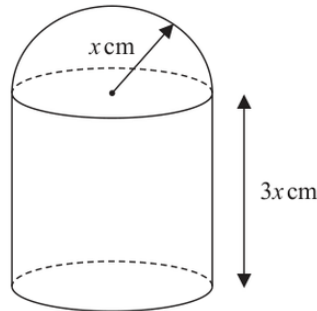
Aluminium

EA

PAST PAPER QUESTION**Q#: 21****Marks: 6****TOPIC: Standard & Compound Units**

Jan 2022 P2HR - Jan 2022 | P2HR

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The metal used to make the solid is one of the metals in the table.

Determine the metal used to make the solid.
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(Total for Question 21 is 6 marks)

PAST PAPER SOLUTION**Q#: 21****Marks: 6****TOPIC: Standard & Compound Units**

Jan 2022 P2HR - Jan 2022 | P2HR

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bottom of cylinder:

$$\pi x^2$$

4. curved surface area of hemisphere

curved surface area of hemisphere:

$$2\pi x^2$$

5. Calculate value

So

$$6\pi x^2 + \pi x^2 + 2\pi x^2 = 81\pi$$

$$9\pi x^2 = 81\pi$$

$$x^2 = 9$$

$$x = 3$$

6. Volume of the cylinder

Volume of the cylinder:

$$\pi x^2(3x) = 3\pi x^3$$

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$$\frac{840}{99\pi} = 2.70\dots$$

10. Compare density

This matches aluminium, whose density is 2.7 g/cm^3 .

FINAL ANSWER

Aluminium

EA

PAST PAPER QUESTION**Q#: 22****Marks: 5****TOPIC: Coordinate Geometry**

Jan 2022 P2HR - Jan 2022 | P2HR

22 ABC is a triangle in which angle $ABC = 90^\circ$ p and q are integers such thatthe coordinates of A are $(p, 10)$ the coordinates of B are $(-1, -5)$ the coordinates of C are $(8, q)$ Given that the gradient of AC is $-\frac{6}{7}$ work out the value of p and the value of q $p = \dots\dots\dots$ $q = \dots\dots\dots$ **(Total for Question 22 is 5 marks)**

PAST PAPER SOLUTION**Q#: 22****Marks: 5****TOPIC: Coordinate Geometry**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. The gradient of is**The gradient of AC is $-\frac{6}{7}$.

$$\frac{q - 10}{8 - p} = -\frac{6}{7}$$

$$7(q - 10) = -6(8 - p)$$

$$7q - 70 = -48 + 6p$$

$$6p - 7q = -22$$

2. Find the gradientSince $\angle ABC = 90^\circ$, lines AB and BC are perpendicular.

$$\text{gradient of } AB = \frac{10 - (-5)}{p - (-1)} = \frac{15}{p + 1}$$

$$\text{gradient of } BC = \frac{q - (-5)}{8 - (-1)} = \frac{q + 5}{9}$$

3. The product of perpendicular gradientsThe product of perpendicular gradients is -1 :

$$\frac{15}{p + 1} \times \frac{q + 5}{9} = -1$$

$$15(q + 5) = -9(p + 1)$$

$$5(q + 5) = -3(p + 1)$$

$$3p + 5q = -28$$

4. Solve

Now solve

$$6p - 7q = -22$$

$$3p + 5q = -28$$

5. Double the second equation

Double the second equation:

$$6p + 10q = -56$$

6. Subtract

Subtract:

$$17q = -34$$

$$q = -2$$

7. Calculate value

Then

$$3p + 5(-2) = -28$$

$$3p = -18$$

$$p = -6$$

FINAL ANSWER

$$p = -6, q = -2$$

EA

PAST PAPER QUESTION**Q#: 22****Marks: 5****TOPIC: Coordinate Geometry**

Jan 2022 P2HR - Jan 2022 | P2HR

22 ABC is a triangle in which angle $ABC = 90^\circ$

p and q are integers such that

the coordinates of A are $(p, 10)$

the coordinates of B are $(-1, -5)$

the coordinates of C are $(8, q)$

Given that the gradient of AC is $-\frac{6}{7}$

work out the value of p and the value of q

$p = \dots\dots\dots$

$q = \dots\dots\dots$

(Total for Question 22 is 5 marks)

PAST PAPER SOLUTION**Q#: 22** **Marks: 5****TOPIC: Coordinate Geometry**

Jan 2022 P2HR - Jan 2022 | P2HR

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Then

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$$3p = -18$$

$$p = -6$$

FINAL ANSWER

$$p = -6, q = -2$$

PAST PAPER QUESTION**Q#: 23****Marks: 4****TOPIC: Functions**

Jan 2022 P2HR - Jan 2022 | P2HR

23 The functions f and g are such that

$$f(x) = x + 25 \qquad g(x) = x^2 - 12x$$

The function h is such that $h(x) = fg(x)$

The domain of h is $\{x : x \leq 6\}$

Express the inverse function h^{-1} in the form $h^{-1}(x) = \dots$

$$h^{-1}(x) = \dots\dots\dots$$

(Total for Question 23 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 23****Marks: 4****TOPIC: Functions**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate value**

$$f(x) = x + 25$$

$$g(x) = x^2 - 12x$$

$$h(x) = fg(x) = f(g(x))$$

$$h(x) = x^2 - 12x + 25$$

$$h(x) = (x - 6)^2 - 11$$

2. Let

Let

$$y = (x - 6)^2 - 11$$

$$y + 11 = (x - 6)^2$$

3. Find inverse functionSince the domain of h is $x < 6$, take the negative square root:

$$x - 6 = -\sqrt{y + 11}$$

$$x = 6 - \sqrt{y + 11}$$

FINAL ANSWER

$$h^{-1}(x) = 6 - \sqrt{x + 11}$$

PAST PAPER QUESTION**Q#: 23****Marks: 4****TOPIC: Functions**

Jan 2022 P2HR - Jan 2022 | P2HR

23 The functions f and g are such that

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The function h is such that $h(x) = fg(x)$

The domain of h is $\{x : x \leq 6\}$

Express the inverse function h^{-1} in the form $h^{-1}(x) = \dots$

$$h^{-1}(x) = \dots\dots\dots$$

(Total for Question 23 is 4 marks)

EA

PAST PAPER SOLUTION**Q#: 23****Marks: 4****TOPIC: Functions**

Jan 2022 P2HR - Jan 2022 | P2HR

WORKED SOLUTION*Dr. Eslam Ahmed***1. Calculate value**

$$f(x) = x + 25$$

$$g(x) = x^2 - 12x$$

$$h(x) = fg(x) = f(g(x))$$

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FINAL ANSWER

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